A SIMPLE ROOT OF A POLYNOMIAL OVER A LOCAL RING IS UNIQUE AMONG THE LIFTS CONGRUENT TO IT MODULO THE MAXIMAL IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance, as required for this source: the first line of the pinned chapter file is \input{preamble}, and the native preamble was not staged with the source, so no preamble text lives inside the chapter. The retained preamble used here is the project's single Stacks-wide preamble (the same 217-line / 8116-byte file that the already published chow and etale-cohomology cards use), copied mechanically line for line from sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Not one character was changed, no package was added, no macro was defined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The same-chapter pointers are located at their \label lines inside the pinned chapter file; the cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The author leaves the Taylor expansion to the reader ('details omitted'); that sentence is kept verbatim inside the counted span and is a routine omission, since the argument that uses the expansion is fully written out.

Editorial note. The staged candidate list records ext_refs = 0 for this label and the counted proof indeed contains no \ref and no \cite, so the card needs no navigation stub and no bibliography.

Allowed standard prerequisites (editorial).

- The Taylor expansion of a polynomial in two variables, f(x+y) = f(x) + f'(x)y + g(x,y)y^2. Used inside the counted proof at its first sentence. Supplies the identity that converts f(b) - f(a) into f'(a)(b-a) plus a multiple of (b-a)^2.
- In a local ring an element of the form 1 + m with m in the maximal ideal is a unit. Used inside the counted proof in its last two sentences. Turns the factor 1 + f'(a)^{-1}c(b-a) into a unit so that the product vanishes only if b - a does.

Lemma 0.1. *Let $(R, \mathfrak{m}, \kappa)$ be a local ring. Let $f \in R[T]$. Let $a, b \in R$ such that $f(a) = f(b) = 0$, $a = b \bmod \mathfrak{m}$, and $f'(a) \notin \mathfrak{m}$. Then $a = b$.*

Proof. Write $f(x+y) - f(x) = f'(x)y + g(x,y)y^2$ in $R[x,y]$ (this is possible as one sees by expanding $f(x+y)$; details omitted). Then we see that $0 = f(b) - f(a) = f(a+(b-a)) - f(a) = f'(a)(b-a) + c(b-a)^2$ for some $c \in R$. By assumption $f'(a)$

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is a unit in R. Hence $(b-a)(1+f'(a)^{-1}c(b-a)) = 0$. By assumption $b-a \in \mathfrak{m}$, hence $1+f'(a)^{-1}c(b-a)$ is a unit in R. Hence $b-a=0$ in R. $\square$